Event Density Geometry of Spin–Momentum Locked Electrons

ED Interpretation of: "The quantum metric of electrons with spin–momentum locking"

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Abstract

Spin–momentum locking in Rashba systems is traditionally described within the quantum-geometric framework, where the quantum metric and Berry curvature are treated as Hilbert-space constructs that modulate electronic response. Recent experiments, however, reveal that the quantum metric behaves as a physically active field: it produces a nonlinear, field-antisymmetric magnetoresistance, diverges at a critical magnetic field, and remains finite even when Berry curvature vanishes. This paper reinterprets these results within the Event Density (ED) ontology, where geometry is not a property of wavefunctions but a property of the event-density flow that constitutes the system. In ED, spin–momentum locking corresponds to a geometric twist of the flow-line bundle, the quantum metric becomes the local deformation tensor describing stretching and shear of that bundle, and the critical field B_c marks a geometric transition where flow-line bundles intersect and reorganize. The nonlinear magnetoresistance emerges as a directional shear bias activated when symmetry constraints shift, while the anomalous planar Hall effect reflects the onset of vorticity in the flow-line field. This reinterpretation unifies metric, curvature, and nonlinear transport as complementary projections of a single geometric structure and predicts new regimes of metric-driven response, disorder-free charge–spin conversion, and geometric engineering of transport in spin–orbit-coupled materials. The result is a coherent geometric account of quantum metric phenomena that positions condensed-matter systems as direct laboratories for ED geometry.

1. Introduction: Geometry as a Physical Actor in ED and in Condensed Matter

The central claim of Event Density (ED) is that geometry is not a background, not a container, and not an abstract mathematical descriptor of wavefunctions. Geometry is what the system does. It is the organization of event density into flow-lines, constraints, and deformation patterns that determine how identities persist, interact, and evolve. In this ontology, curvature, torsion, and metric structure are not properties of a Hilbert space or a manifold but signatures of how the underlying event-density field organizes itself.

Condensed-matter physics has been converging—quietly but unmistakably—toward this same insight. Over the past decade, the field has shifted from band structures and quasiparticles toward geometric quantities: Berry curvature, Berry curvature dipoles, quantum geometric tensors, and the quantum metric. These objects were originally introduced as mathematical constructs, but they increasingly behave like real geometric fields that shape transport, optical response, and nonlinear phenomena.

The paper "The quantum metric of electrons with spin-momentum locking" provides one of the clearest demonstrations of this shift. It shows that a finite quantum metric emerges generically in systems with spin–momentum locking, even when Berry curvature vanishes. It shows that this metric produces a nonlinear magnetoresistance that can be electrically tuned. And it shows that the metric behaves as an intrinsic geometric property of the electronic system, independent of scattering and disorder.

This is precisely the kind of phenomenon ED predicts: geometry emerging from the organization of flow-lines, not from the algebra of wavefunctions. The quantum metric, in ED terms, is not a Hilbert-space distance measure

but a real deformation tensor describing how the event-density flow responds to displacement. The nonlinear magnetoresistance is not a higher-order correction but a geometric bias activated when symmetry constraints shift. And the divergence at the critical field B_c is not a transport anomaly but a geometric transition point where flow-line bundles intersect and reorganize.

The purpose of this reinterpretation paper is to translate the full content of the original work into ED's geometric ontology. The goal is not to critique the conventional account but to reveal the deeper geometric structure that ED makes explicit. By doing so, we show that the quantum metric is not an exotic or rare quantity but a natural and inevitable feature of systems whose event-density flow is twisted, stretched, or sheared by spin–orbit coupling. We show that nonlinear magnetoresistance is a geometric response, not a perturbative one. And we show that condensed-matter systems with spin–momentum locking provide a clean, experimentally accessible platform for probing ED geometry directly.

This reinterpretation also serves a broader purpose. It demonstrates that ED is not limited to high-energy physics, cosmology, or quantum foundations. It applies equally well to the geometry of electrons in solids, to nonlinear transport, and to the quantum geometric tensor. Condensed-matter systems become laboratories for ED geometry, and ED becomes a unifying language for phenomena that currently appear fragmented across topology, nonlinear optics, and spintronics.

The remainder of this paper reconstructs the conventional account, identifies the conceptual gaps that ED resolves, and then provides a full ED reinterpretation of the quantum metric, the BNQM dipole, the critical field B_c, and the nonlinear magnetoresistance observed in 111-oriented $LaAlO_3/SrTiO_3$ interfaces.

2. Quantum Summary of the Original Paper

This section reconstructs the mainstream interpretation of the results, using the language and assumptions of condensed-matter physics. It provides the baseline against which the ED reinterpretation will be contrasted.

2.1 Spin–Momentum Locking and Rashba Bands

The system under study is a two-dimensional electron gas with strong spin–orbit coupling, described by a Rashba-type Hamiltonian. The Rashba term shifts the parabolic bands in momentum space and produces helical spin textures. As the authors note:

"The linear coupling between the crystalline momentum k and the electron spins… leads to helical spin textures… characterized by the complete absence of the Berry curvature."

This is a crucial point: in time-reversal symmetric Rashba systems, Berry curvature vanishes identically. Yet the system exhibits geometric structure, suggesting that Berry curvature is not the only geometric quantity of interest.

2.2 Quantum Metric and the Band-Normalized Quantum Metric (BNQM)

The quantum geometric tensor G has a real part (the quantum metric g) and an imaginary part (the Berry curvature Ω). The quantum metric measures the "distance" between nearby Bloch states in Hilbert space. The authors emphasize:

"The quantum metric g corresponds to the real part of the quantum geometric tensor… the BNQM is finite in the region of momenta populated by a single spin band."

Even when Berry curvature is zero, the quantum metric remains finite. The BNQM dipole density A_{xxx} and A_{yyy} exhibit a quadrupolar pattern in momentum space, but their integrated values vanish in time-reversal symmetric conditions.

2.3 Magnetic-Field Activation of Nonlinear Magnetoresistance

A planar magnetic field breaks time-reversal symmetry and shifts the singularities and zero-lines of the quantum metric. This destroys the quadrupolar symmetry of the BNQM dipole and produces a finite integrated dipole component. The authors write:

"The planar magnetic field activates a nonlinear in-plane magnetoresistance… which becomes finite when the BNQM dipole loses its quadrupolar profile."

This effect is called quantum metric magnetoresistance (QMMR). It is distinct from the bilinear magnetoelectric resistance (BMER), which requires different symmetry conditions.

2.4 Experimental Platform: 111-Oriented $LaAlO_3/SrTiO_3$

The authors use a 2D electron gas at the 111-oriented $LaAlO_3/SrTiO_3$ interface. This system hosts three Kramers pairs of Rashba-split bands. Gate voltage tunes the carrier density and can induce a Lifshitz transition. The Rashba effect endows all bands with a finite quantum metric, but the nonlinear response is dominated by the lowest-filled band because its critical field B_c is experimentally accessible.

2.5 Scaling Analysis and Disentangling QMMR from BMER

The key methodological advance is the scaling analysis. QMMR is intrinsic and scales as τ^0 independent of scattering time), while BMER scales as τ^2. By analyzing the dependence of nonlinear conductivity on mobility, the authors separate the two contributions. They find:

"The nonlinear in-plane magnetoresistance… features a quantum metric contribution that is three orders of magnitude larger than that reported for 2D topological antiferromagnets."

This establishes the quantum metric as a dominant geometric actor in nonlinear transport.

3. The Missing Mechanism in the quantum Account

The conventional interpretation of the quantum metric in Rashba systems treats geometry as a property of wavefunctions rather than a property of the physical system. This framing is mathematically consistent but conceptually incomplete. The experimental results in the paper expose several behaviors that do not sit comfortably inside the Hilbert-space picture but fall out naturally and inevitably in the ED ontology. This section identifies those gaps.

Geometry is treated as an abstract descriptor, not a physical actor

In the standard view, the quantum metric is defined as the real part of the quantum geometric tensor, a measure of the "distance" between nearby Bloch states. This distance is not spatial, not dynamical, and not tied to any physical flow. It is a purely mathematical construct. Yet the experiments show that this metric produces a real,

measurable nonlinear current, tunable by magnetic field and gate voltage. The metric behaves like a physical field, not an abstract one.

This mismatch is the first conceptual gap: the quantum account cannot explain why a Hilbert-space distance measure should act like a real geometric force.

The quantum metric survives when Berry curvature vanishes
In Rashba systems with time-reversal symmetry, Berry curvature is identically zero. The standard picture interprets this as "no geometric field," because Berry curvature is the quantity normally associated with geometric transport. Yet the quantum metric remains finite and produces a nonlinear response.

The paper states:

"The quantum metric… is generally finite… even in the complete absence of Berry curvature."

This is a second conceptual gap: the quantum account has no physical mechanism for geometry without curvature. In ED, this distinction is natural—stretching and shearing of flow-lines can exist without vorticity—but the conventional picture lacks such a decomposition.

The BNQM dipole reorganizes under magnetic field in a way that resembles flow-line shear
The BNQM dipole has a quadrupolar structure in momentum space. When a planar magnetic field is applied, the quadrupole collapses into a unidirectional pattern, producing a finite integrated dipole and a nonlinear current. The paper describes this as:

"The BNQM dipoles lose their quadrupolar profile… the integrated dipole becomes finite."

This behavior is geometric in the physical sense: a symmetric deformation pattern becomes asymmetric under an external field. But the quantum account treats this as a change in the distribution of Hilbert-space distances, not as a deformation of anything physical.

This is the third conceptual gap: the observed reorganization behaves like a real geometric shear, but the conventional interpretation has no physical geometry to shear.

The divergence at the critical field B_c has no physical explanation
The nonlinear response diverges at a critical magnetic field where the two Rashba-split Fermi lines intersect. The paper notes:

"Its magnitude increases linearly with B… and diverges at the critical field B_c."

In the standard picture, this divergence is a mathematical feature of the BNQM dipole. But there is no physical mechanism explaining why a Hilbert-space metric should blow up at a band-touching point. In ED, this is a natural geometric transition: flow-line bundles intersect, and the deformation tensor becomes singular.

This is the fourth conceptual gap: the quantum account predicts the divergence but cannot explain it.

The scaling behavior (τ^0 vs τ^2) reveals intrinsic geometry

The authors disentangle QMMR from BMER using mobility scaling. QMMR is intrinsic and independent of scattering time, while BMER scales with τ^2. The paper states:

"The QMMR contribution is intrinsic in origin and scales as τ^0."

This is a hallmark of a geometric effect: it does not depend on scattering, impurities, or disorder. But in the conventional picture, the quantum metric is not a physical field, so its independence from scattering is conceptually unanchored.

This is the fifth conceptual gap: the quantum account identifies intrinsic behavior but cannot explain why the metric should be intrinsic.

The quantum picture cannot unify metric, curvature, and nonlinear response
The paper extracts three contributions—quantum metric, Berry curvature, and BMER—and treats them as separate mechanisms. But the experimental data show that they are intertwined, with similar field dependence and shared symmetry constraints.

The authors write:

"The intrinsic contributions… provide direct access to the Berry curvature and the quantum metric… and allow us to probe the full quantum geometric tensor."

The quantum account treats these as separate geometric quantities with separate physical roles. ED treats them as different projections of the same underlying event-density geometry.

This is the sixth conceptual gap: the quantum picture fragments what ED unifies.

4. ED Reinterpretation: Quantum Metric as Event-Density Geometry

The standard quantum account treats the quantum metric as a Hilbert-space object: a measure of the infinitesimal distance between Bloch states. It is defined on the manifold of wavefunctions, not on physical space, and is interpreted as a geometric correction to electronic response. In ED, this entire framing is inverted. Geometry is not a property of wavefunctions; it is a property of the event-density field that constitutes the system. The quantum metric is not an abstract distance measure but a real deformation tensor describing how the event-density flow-lines stretch, shear, and reorganize under constraints.

This section reconstructs the entire phenomenon—spin–momentum locking, BNQM dipoles, the critical field B_c, and nonlinear magnetoresistance—as consequences of ED geometry.

4.1 Geometry in ED: Flow-Lines, Constraints, and Deformation

In ED, every physical system is a structured flow of event density.

- Flow-lines represent persistent identity.
- Constraints (spin–orbit coupling, symmetry, external fields) shape the flow.
- Geometry is the pattern of deformation in the flow-line bundle.

The ED metric is not a background metric; it is the local response of the flow-line geometry to displacement. Stretching, shearing, and compression are physical, not mathematical. When the flow-line field is twisted or anisotropically constrained, the ED metric becomes nontrivial.

This is the ED counterpart of the quantum metric.

4.2 Spin–Momentum Locking as a Geometric Twist
In the standard quantum account, spin–momentum locking arises from the Rashba Hamiltonian:

$$H_R = \alpha_R(\hat{z} \times k) \cdot \sigma.$$

In ED, this is not a coupling term but a geometric twist in the event-density flow.

- Momentum direction determines spin orientation.
- Spin orientation determines how the flow-lines can bend.
- The result is a helical flow-line texture.

This helical structure is a real geometric configuration of the event-density field. It is not surprising, in ED, that such a twist produces a finite metric even when curvature (vorticity) is zero. Twisting without vorticity is a standard geometric configuration in ED.

This explains the paper's central observation:

"The quantum metric… is generally finite… even in the complete absence of Berry curvature."

In ED, this is expected: curvature = vorticity; metric = stretching/shearing.
You can twist a bundle without giving it vorticity.

4.3 The Quantum Metric as a Deformation Tensor of the Flow-Line Bundle
The quantum metric g_{ij} measures the sensitivity of Bloch states to momentum displacement. In ED, this becomes:

The local deformation tensor describing how the event-density flow-lines stretch when displaced in momentum space.

The BNQM dipole A_{xxx}, A_{yyy} corresponds to:

The anisotropic shear pattern of the flow-line bundle.

The quadrupolar structure seen in Fig. 1C of the paper is the ED signature of a balanced shear field: equal positive and negative deformation regions arranged symmetrically.

The singular point at $k = 0$ is a topological defect in the flow-line geometry.

The lines where g_{xx} or g_{yy} vanish are neutral shear lines where the flow-line bundle is locally rigid.

All of these features are geometric in the ED sense, not algebraic.

4.4 Why Berry Curvature Vanishes but the Metric Does Not

In the standard quantum account, this is puzzling:

- Berry curvature = 0
- Quantum metric ≠ 0
- Yet nonlinear responses appear.

In ED, the distinction is natural:

- Berry curvature corresponds to vorticity of the flow-lines.
- Quantum metric corresponds to stretching/shearing of the flow-lines.

A twisted but non-vortical flow has:

- zero curvature,
- finite metric.

Rashba systems are exactly this: twisted but not swirling.

This resolves the conceptual gap in the standard quantum account.

4.5 Magnetic-Field Activation as Geometric Unbalancing

A planar magnetic field shifts the spin textures and moves the singular point of the metric. In ED, this is a geometric unbalancing of the flow-line bundle.

Before the field:

- The shear pattern is quadrupolar.
- Positive and negative regions cancel.
- No net geometric bias → no nonlinear current.

After the field:

- The quadrupole collapses.
- The shear pattern becomes unidirectional.
- A net geometric bias appears → nonlinear current.

This is the ED interpretation of the BNQM dipole becoming finite.

The paper describes this as:

"The BNQM dipoles lose their quadrupolar profile… the integrated dipole becomes finite."

In ED, this is simply:
the flow-line bundle becomes directionally sheared.

4.6 The Critical Field B_c as a Geometric Transition Point

The divergence at B_c is one of the most striking features of the experiment. In the standard quantum account, it is a mathematical divergence of the BNQM dipole when the Fermi lines intersect.

In ED, this is a geometric transition:

- At B_c, two flow-line bundles intersect at a degeneracy point.
- The deformation tensor becomes singular because the flow-lines cannot maintain their previous configuration.
- The system undergoes a topological reorganization.

This is the ED analogue of a caustic or fold catastrophe in geometric optics.

The divergence is not a transport anomaly; it is a geometric instability.

4.7 Why QMMR Is Intrinsic (τ^0 Scaling)
The paper finds:

"The QMMR contribution is intrinsic in origin and scales as τ^0."

In ED, this is required:

- Geometry is intrinsic.
- Scattering affects only extrinsic responses.
- A geometric deformation tensor cannot depend on impurity scattering.

The τ^0 scaling is direct evidence that the quantum metric is a real geometric field, not a perturbative correction.

4.8 Unifying Metric, Curvature, and Nonlinear Response
The standard quantum account treats:

- quantum metric,
- Berry curvature,
- BMER

as separate mechanisms.

In ED, they are different projections of the same underlying geometry:

- Metric = stretching/shearing
- Curvature = vorticity
- BMER = mixed geometric–scattering response
- QMMR = pure geometric bias

The full quantum geometric tensor becomes the ED flow-line geometry tensor.

This unifies the entire phenomenon.

5. ED Interpretation of the Experimental Results

The experimental results in the paper become far more transparent when viewed through ED's geometric ontology. Instead of interpreting the data as signatures of Hilbert-space geometry, ED treats them as direct probes of how the event-density flow-lines deform, shear, and reorganize under symmetry constraints, gating, and magnetic fields. Each experimental observation corresponds to a geometric transition or reconfiguration in the underlying flow-line bundle.

5.1 Gate-Tuned Lifshitz Transition as a Geometric Phase Change

In the standard quantum account, the Lifshitz transition marks the point where the chemical potential crosses into a new Rashba-split band, activating additional carriers. In ED, this is a geometric phase transition in the event-density field.

- Each Kramers pair corresponds to a distinct flow-line bundle with its own twist, shear, and rigidity.
- When the gate voltage raises the chemical potential, a new bundle becomes populated.
- The onset of nonlinear magnetoresistance coincides with the activation of this new geometric channel.

The paper notes that the nonlinear response emerges sharply when the second Kramers pair becomes populated:

"The enhancement of the nonlinear response practically coincides with the filling of the second pair of Kramers bands."

In ED terms, this is the moment when a new flow-line bundle begins to participate in the deformation geometry. The system acquires a new shear mode, and the nonlinear response becomes possible.

5.2 Mobility Scaling as Evidence for Intrinsic Geometry

The authors disentangle QMMR from BMER using mobility scaling. QMMR is intrinsic and independent of scattering time, while BMER scales with τ^2. This is a decisive signature of geometric origin.

In ED:

- Intrinsic geometry (flow-line deformation) must be independent of scattering.
- Extrinsic responses (skew scattering, side-jump) must depend on scattering time.

The paper states:

"The QMMR contribution is intrinsic in origin and scales as τ^0."

This is exactly what ED predicts. A geometric deformation tensor cannot depend on impurity scattering because it is a property of the flow-line field itself. The τ^0 scaling is not just consistent with ED — it is diagnostic of ED geometry.

5.3 Anomalous Planar Hall Effect as Curvature Activation

The anomalous planar Hall effect (APHE) appears when the magnetic field reaches a threshold and the avoided crossing at B_c enters the Fermi surface annulus. In the standard quantum account, this is attributed to Berry curvature hotspots activated by symmetry breaking.

In ED, this corresponds to the activation of vorticity in the flow-line bundle.

Before the threshold, the flow-lines are twisted but not vortical.

When the field shifts the degeneracy point into the populated region, the flow-lines acquire local vorticity.

This produces a transverse geometric bias → APHE.

The paper describes this as:

"The appearance of the anomalous planar Hall effect… confirms that the critical magnetic field B_c is experimentally accessible."

In ED, this is the moment when the flow-line geometry transitions from pure shear to mixed shear-vorticity.

5.4 Nonlinear Magnetoresistance as Geometry-Activated Flow Bias
The nonlinear in-plane magnetoresistance is the central experimental signature of the quantum metric. In ED, this is the direct manifestation of directional shear in the flow-line bundle.

- Before the magnetic field: the shear pattern is quadrupolar → no net bias.
- After the field: the quadrupole collapses → net shear → nonlinear current.

The paper reports:

"A nonlinear and field-antisymmetric in-plane magnetoresistance… sharply emerges at a threshold conductivity."

In ED, this threshold corresponds to the moment when the flow-line geometry becomes sufficiently asymmetric to produce a net directional bias. The nonlinear current is not a perturbative correction — it is the macroscopic expression of a geometric asymmetry.

5.5 Divergence at the Critical Field B_c as a Geometric Instability
The divergence of the nonlinear response at B_c is one of the most striking features of the experiment. In the standard quantum account, this is a mathematical divergence of the BNQM dipole when the Fermi lines intersect.

In ED, this is a geometric instability:

- At B_c, two flow-line bundles intersect at a degeneracy point.
- The deformation tensor becomes singular because the flow-lines cannot maintain their previous configuration.
- The system undergoes a topological reorganization.

This is analogous to a caustic in optics or a fold catastrophe in dynamical systems. The divergence is not a transport anomaly — it is the signature of a geometric transition.

5.6 Full Quantum Geometric Tensor as the ED Flow-Line Geometry Tensor
The paper extracts three contributions:

- quantum metric (intrinsic, τ^0),
- Berry curvature (intrinsic, symmetry-dependent),
- BMER (extrinsic, τ^2).

In the standard quantum account, these are separate mechanisms. In ED, they are unified:

- Metric = stretching/shearing of flow-lines
- Curvature = vorticity of flow-lines
- BMER = scattering-modulated geometric response
- QMMR = pure geometric bias

The full quantum geometric tensor becomes the ED flow-line geometry tensor, with metric and curvature as complementary projections of the same underlying structure.

The paper's statement:

"The intrinsic contributions… allow us to probe electronic dynamics associated with the full quantum geometric tensor."

is, in ED terms:

The experiment directly probes the geometry of the event-density field.

5.7 Summary: The Experiments Are Direct Probes of ED Geometry
Every experimental observation — the onset of nonlinear response, the divergence at B_c, the mobility scaling, the APHE, the gate-tuned transitions — corresponds to a geometric transition or deformation in the ED flow-line field.

The standard quantum account identifies the patterns but cannot explain their physical origin. ED provides the missing mechanism: geometry as a physical actor, not a mathematical descriptor.

6. Consequences for ED Geometry and Condensed-Matter Physics

The reinterpretation of the quantum metric as a real geometric property of the event-density field has several deep consequences. These consequences are not cosmetic; they reshape how condensed-matter systems are conceptualized, how nonlinear transport is interpreted, and how new materials and devices might be engineered. The results of the paper become not just an isolated observation but a demonstration that ED geometry is experimentally accessible, tunable, and technologically relevant.

6.1 Quantum Metric Effects Become Ubiquitous, Not Exceptional
The standard quantum account treats the quantum metric as a subtle, often inaccessible quantity that appears only in special systems—topological antiferromagnets, flat bands, or engineered photonic structures. The paper already hints that this is too restrictive:

"The quantum metric is… not contingent on specific symmetry requirements beside the broken inversion symmetry."

In ED, this becomes even stronger:

- Any system with spin–momentum locking has a twisted flow-line geometry.
- Any twisted geometry has finite metric (stretching/shearing).
- Therefore, quantum metric effects are generic, not rare.

This reframes the entire field. Instead of searching for exotic materials where the quantum metric might be detectable, ED predicts that most spin–orbit-coupled materials already host strong geometric effects—they simply haven't been interpreted that way.

This includes:

- oxide interfaces,
- semiconductor heterostructures,
- heavy-metal surfaces,
- Rashba interfaces,
- transition-metal dichalcogenides,
- even polycrystalline samples.

The quantum metric becomes a universal geometric actor.

6.2 ED Predicts New Nonlinear Responses Beyond QMMR

The experiment demonstrates one nonlinear response: quantum metric magnetoresistance (QMMR). But ED geometry implies a much broader family of nonlinear effects, because any geometric deformation of the flow-line field can produce directional biases.

ED predicts:

- nonlinear Hall-like responses without Berry curvature,
- nonlinear spin-orbit torques driven by shear rather than vorticity,
- THz photocurrents activated by geometric asymmetry,
- gate-controlled geometric switching between shear-dominated and vorticity-dominated regimes,
- nonlinear optical rectification tied to flow-line deformation.

The paper hints at this when it notes:

"The quantum metric… is, in principle, accessible to THz photocurrents and spectroscopic tools."

In ED, this is not a possibility—it is an inevitability. Any probe that couples to flow-line geometry will detect metric-driven nonlinearities.

6.3 ED Predicts Disorder-Free Charge–Spin Conversion

One of the most important technological implications is that ED geometry provides a disorder-free mechanism for converting charge currents into spin-orbital currents.

The paper states:

“Combining spintronics with the quantum metric may reveal unforeseen disorder-free mechanisms to interconvert charge and spin-orbital currents.”

In ED, this is straightforward:

- A directional shear in the flow-line bundle biases the spin texture.
- A charge current moving through a sheared geometry acquires a spin-orbital component.
- No scattering is required; the geometry itself performs the conversion.

This is a fundamentally different mechanism from spin Hall or Rashba–Edelstein effects, which rely on curvature or scattering. ED predicts a new class of metric-driven spin-orbitronics.

6.4 ED Predicts Geometric Engineering of Transport
Because the quantum metric is a deformation tensor of the event-density field, ED predicts that it can be engineered by shaping the geometry of the system.

Possible geometric controls include:

- strain (modifies flow-line rigidity),
- interface orientation (changes twist pattern),
- patterned magnetic fields (shift singularities),
- electrostatic gating (activates new flow-line bundles),
- layer stacking (introduces interlayer shear modes).

The paper hints at this when it notes that integrating spin–momentum locking with magnetic or ferroelectric films could enable reversible control of quantum metric effects.

In ED, this becomes a design principle:

- geometry is a tunable resource,
- flow-line deformation is a functional degree of freedom,
- transport can be engineered by shaping the event-density field.

This opens the door to “geometric electronics,” where devices are designed not by band engineering but by flow-line geometry engineering.

6.5 ED Unifies Metric, Curvature, and Nonlinear Transport
The standard quantum account treats:

- quantum metric
- Berry curvature,
- BMER,
- nonlinear Hall effects,
- nonlinear magnetoresistance

as separate mechanisms with separate symmetry requirements.

In ED, they are unified:

- Metric = stretching/shearing
- Curvature = vorticity
- BMER = scattering-modulated geometry
- QMMR = pure geometric bias
- APHE = vorticity activation
- Nonlinear optical responses = geometric asymmetry under oscillatory drive

This unification is not aesthetic—it is predictive. It means that:

- whenever curvature appears, metric must also appear;
- whenever metric appears, nonlinear responses must be possible;
- whenever symmetry is broken, geometric biases must emerge.

The paper's extraction of the full quantum geometric tensor becomes, in ED, the first experimental measurement of the flow-line geometry tensor.

6.6 ED Predicts New Regimes of Quantum Geometry in Multiband Systems
The experiment shows that the nonlinear response is dominated by the lowest-filled Rashba band because its critical field B_c is accessible. In ED, this is a statement about flow-line hierarchy.

ED predicts:

- each band corresponds to a distinct geometric mode,
- each mode has its own critical field, shear pattern, and vorticity structure,
- multiband systems host interference between geometric modes,
- gating can activate or deactivate geometric channels.

This leads to new predictions:

- geometric resonances when two flow-line bundles interact,
- geometric transparency windows where shear cancels,
- geometric amplification near multiband degeneracies.

These are not visible in the standard quantum account but follow directly from ED geometry.

6.7 ED Reframes Condensed Matter as a Geometry Laboratory
The most profound consequence is conceptual. Condensed-matter systems become laboratories for ED geometry. Instead of relying on cosmology, high-energy physics, or quantum foundations, ED can be tested, refined, and expanded using tabletop experiments.

The paper demonstrates:

- geometric fields (metric, curvature) can be measured,
- geometric transitions (at B_c) can be induced,
- geometric asymmetries can produce macroscopic currents,

- geometric tensors can be extracted from transport data.

This is exactly what ED needs: a domain where geometry is tunable, observable, and technologically relevant.

7. Conclusion

The reinterpretation of the quantum metric through the lens of ED geometry reveals a coherent and physically grounded picture of nonlinear transport in spin–orbit-coupled systems. The standard quantum account identifies the mathematical structures—quantum metric, Berry curvature, BNQM dipoles—but leaves their physical origin opaque. ED provides the missing mechanism: geometry as a real, dynamical property of the event-density field.

This reinterpretation explains why the quantum metric remains finite when Berry curvature vanishes, why nonlinear magnetoresistance emerges only when symmetry is broken, why the response diverges at the critical field B_c, and why the intrinsic contribution is independent of scattering. It unifies metric, curvature, and nonlinear transport as different projections of the same underlying geometric structure.

The consequences extend beyond this specific experiment. ED predicts that quantum metric effects are ubiquitous in spin–orbit-coupled materials, that nonlinear responses can be engineered through geometric control, and that condensed-matter systems provide a powerful platform for probing ED geometry directly. The quantum metric becomes not a mathematical curiosity but a functional degree of freedom, enabling new mechanisms for charge–spin conversion, nonlinear optics, and geometric electronics.

This paper demonstrates that ED is not confined to foundational physics or cosmology. It applies equally well to the geometry of electrons in solids, offering a unified and physically intuitive framework for understanding the quantum geometric tensor and its experimental signatures. The result is a deeper, more coherent picture of quantum geometry—one in which the metric, curvature, and nonlinear responses arise naturally from the structure of the event-density field.

References

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